FEATURE ENCODING & SCALING

AI & ML Internship – Task 4 Report

1. Introduction

Feature Encoding and Scaling are important preprocessing techniques in Machine Learning. This task focuses on converting categorical data into numerical form and scaling numerical features for better machine learning model performance using Python libraries.

2. Objective

• To identify categorical columns
• To apply Label Encoding
• To apply One Hot Encoding
• To apply StandardScaler and MinMaxScaler
• To prepare the dataset for machine learning algorithms

3. Dataset Information

Dataset Used: Adult Income Dataset
File Format: CSV
Target Column: Income

4. Technologies Used

Python
Pandas
NumPy
Scikit-learn
Jupyter Notebook / VS Code

5. Steps Performed

Step 1: Imported Required Libraries
Step 2: Loaded Adult Income Dataset
Step 3: Renamed Target Column
Step 4: Checked Dataset Information
Step 5: Checked Missing Values
Step 6: Identified Categorical Columns
Step 7: Applied Label Encoding
Step 8: Applied One Hot Encoding
Step 9: Applied StandardScaler
Step 10: Applied MinMaxScaler

6. Output

• Categorical columns converted into numerical form
• One Hot Encoding applied successfully
• Feature scaling completed successfully
• Dataset transformed into machine learning ready format

7. Conclusion

In this task, categorical data was converted into numerical form using Label Encoding and One Hot Encoding. Feature scaling techniques such as StandardScaler and MinMaxScaler were applied successfully. The dataset was transformed into a machine learning ready format for further analysis and model building.